

CHAMELI DEVI GROUP OF INSTITUTIONS INDORE (M.P.)



**Name of Mini Project: VISA APPLICATION PROCESSING
SYSTEM**

Mini Project Report

Database Management System

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DEPARTMENT OF INFORMATION TECHNOLOGY

CHAMELI DEVI GROUP OF INSTITUTIONS INDORE (M.P.)



DEPARTMENT OF INFORMATION TECHNOLOGY CERTIFICATE

This is to certify that Mr./Ms. **Asha Porwal, Harshita Baghel, Himanshi Joshi, Rashmi Kanojia** with RGTU Enrollment No **0832IT221007, 0832IT221023, 0832IT221025, 0832IT221049**, has satisfactorily completed the Mini Project on “**VISA APPLICATION PROCESSING SYSTEM**” in “**Database Management System**”, for **B. Tech, IV Semester** of the **Information Technology Department** during year **2023-24**.

Prof. Radheshyam Acholiya
Head of the Department

Signature of
Faculty In-charge

TABLE OF CONTENTS

CONTENTS	PAGE NO.
Introduction of the Project	04
Description of the Project	05
Technical Details of the Project	06-07
Screenshot of the Code (Min. 3)	08-11
Screenshot of working Project (Min. 5)	12-17
References	18

Introduction of the Project

This Visa Processing System will speed up the time required to get the visa while the software is mainly used for applying for the visa also, it will store the information for the future transaction. Hence this software will be helpful to get the visa and the information related to visa in reduced time.

In the Visa Processing System which is present, works on a manual basis and it does not provide an effective visa processing system. As the process is manual there are a lot of registers present, which decrease the accuracy in the information to be stored and increases the cost to store the details while the time required to access the information also increased.

The database is developed using the Mysql and workbench basic game mechanics such as keyboard input, Queries.

This system will provide a user-friendly platform for applicants to submit their visa applications online, track their application status in real-time, and receive automated notifications about their application progress.

Additionally, it will offer streamlined tools for visa processing authorities to manage applications efficiently, including document verification, decision-making, and communication with applicants. The system's ultimate goal is to enhance transparency, speed, and accuracy in the visa application process, benefiting both applicants and processing authorities.

Description of the Project

This module will maintain the details of applied visa, passport, the number of times any particular man has visited on the site, the number of tours has done by any person.

The admin will be a manager or HR executive of the organization, it will have the control over the number of users who can access the system. The admin can apply for visa renewal and can access the details regarding visa approval.

This contains the details regarding the date of issues of visa, the date of flight, for how many days visa has issued.

This will provide the information related to the person who went to abroad, the progress in the work, the status of visa.

The "Visa Application Processing System" project is designed to modernize and streamline the visa application process. By leveraging digital technologies, the system aims to simplify application submission, enhance tracking capabilities for applicants, and improve efficiency for processing authorities. Through automated notifications and centralized data management, the system will ensure smoother communication and faster decision-making. Ultimately, this initiative seeks to create a more user-friendly and efficient visa application experience for all stakeholders involved.

Technical Details of the Project

The Visa Application Processing System employs MySQL as its primary database management system. This choice is driven by MySQL's reputation for reliability, performance, and scalability in handling large datasets and concurrent user interactions.

The system's database schema is carefully crafted to efficiently store and manage various types of data, including applicant information, visa types, documents, and application statuses. The schema design follows best practices of normalization to minimize data redundancy and ensure data integrity.

Data storage within MySQL is organized into tables, each representing a specific entity or aspect of the visa application process. Relationships between these entities are established through foreign key constraints, facilitating data consistency and integrity.

Indexing is utilized to optimize query performance, with indexes strategically created on frequently accessed fields. Data migration between different environments, such as development, staging, and production, is seamlessly handled using MySQL's native tools and custom scripts.

Hardware requirements:

Great! With a Windows 10 64-bit system equipped with an Intel Core i7 processor and 64 GB of RAM, you have a powerful setup that is well-suited for running MySQL efficiently, especially for larger databases or demanding workloads. Here's how this hardware configuration enhances your MySQL usage:

➤ Hardware Configuration:

- Processor: Intel Core i7
- Clock Speed: Provides high processing power with multiple cores and threads, ideal for handling complex database queries and operations efficiently.
- RAM: 64 GB

- Benefits: Ample memory capacity ensures that MySQL can cache data effectively, improving performance by reducing disk I/O and supporting a large number of concurrent connections and transactions.
- Storage:
 - SSD: Consider using SSD storage for MySQL data and log files, enhancing read/write speeds and overall database responsiveness.
- Network:
 - Ethernet: Utilize a high-speed Ethernet connection for fast data transfer between the database server and client applications.
- Software Recommendations:
 - MySQL Server:
 - Install the latest version of MySQL Server (e.g., MySQL 8.0) to leverage performance optimizations and new features.
 - Windows 10 64-bit:
 - Ensure all Windows updates are installed to maintain system security and compatibility with MySQL Server.

Screenshot of the Code (Min. 3)

The screenshot displays the MySQL Workbench interface. The top menu bar includes File, Edit, View, Query, Database, Server, Tools, Scripting, and Help. The left sidebar shows the 'Schemas' tab with a tree view of databases: employees, sys, visa, and visaa. The 'visa' database is selected, and its contents (Tables, Views, Stored Procedures, Functions) are visible. The main editor window shows the following SQL code:

```
1 • show databases;
2
3 • CREATE DATABASE visa;
4
5 • show databases;
6
7 • USE visa;
8
9 • CREATE TABLE user
10 (user_name VARCHAR(50) NOT NULL,
11   user_contact VARCHAR(20) NOT NULL,
12   user_address VARCHAR(255) NOT NULL,
13   user_id INT NOT NULL,
14   PassportNumber VARCHAR(20),
15   Nationality VARCHAR(50),
16   DateOfBirth DATE,
17   Gender VARCHAR(10),
18   Email VARCHAR(100),
19   PRIMARY KEY (user_id));
```

The bottom panel shows the 'Output' tab with 'Action Output' selected. It contains a table with columns: #, Time, Action, Message, and Duration / Fetch. The table is currently empty.

The Windows taskbar at the bottom shows the system clock as 4:07 PM on 5/12/2024, along with weather information (103°F, Mostly sunny) and various application icons.

MySQL Workbench

DBMS Project x

File Edit View Query Database Server Tools Scripting Help

Navigator

SCHEMAS

Filter objects

- employees
- sys
- visa
 - Tables
 - Views
 - Stored Procedures
 - Functions
- visaa

Administration Schemas

Information

No object selected

Limit to 2000 rows

```

43      ('Henry Gill', '8856934245', '56 Main Road', 128, 'FDC453', 'USA', '1999-08-06', 'Male', 'henry6@example.com');
44
45 • SELECT * FROM user;
46
47
48 • CREATE TABLE visa_application
49     (user_id INT NOT NULL,
50      appl_id VARCHAR(28) NOT NULL,
51      visa_country VARCHAR(50) NOT NULL,
52      visa_type VARCHAR(30) NOT NULL,
53      user_passport_no VARCHAR(20) NOT NULL,
54      status VARCHAR(20), -- Pending, Approved, Rejected, etc.
55      PRIMARY KEY(appl_id),
56      FOREIGN KEY (user_id) REFERENCES user(user_id)
57     );
58
59 • desc visa_application;
60
61 • INSERT INTO visa_application(user_id, appl_id, visa_country, visa_type, user_passport_no, status)
62     VALUES(101, 'PR100101', 'INDIA', 'Education', 'SJU427', 'Approved'),
63           (102, 'PR100102', 'INDIA', 'work', 'LRV396', 'Pending'),
64           (103, 'PR100103', 'INDIA', 'Education', 'KGF684', 'Approved'),
65           (104, 'PR100104', 'UK', 'work', 'JFK320', 'Approved'),
66           (105, 'PR100105', 'India', 'Tourist', 'BVC154', 'Rejected'),

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
---	------	--------	---------	------------------

Object Info Session

103°F Mostly sunny

Search

ENG IN

4:07 PM 5/12/2024

MySQL Workbench

DBMS Project x

File Edit View Query Database Server Tools Scripting Help

Navigator

Limit to 2000 rows

SCHEMAS

Filter objects

- employees
- sys
- visa
 - Tables
 - Views
 - Stored Procedures
 - Functions
- visaa

Administration Schemas

Information

No object selected

```
82
83 • SELECT * FROM visa_application;
84
85 • CREATE TABLE ProcessingStages (
86     StageID INT PRIMARY KEY,
87     StageName VARCHAR(100));
88
89 • desc ProcessingStages;
90
91 • INSERT INTO ProcessingStages (StageID, StageName)
92     VALUES
93     (1, 'Document Verification'),
94     (2, 'Interview'),
95     (3, 'Background Check');
96
97 • SELECT * FROM ProcessingStages;
98
99 • CREATE TABLE ProcessingHistory
100     (HistoryID INT PRIMARY KEY,
101     appl_id VARCHAR(20) NOT NULL,
102     StageID INT,
103     StartDate DATE,
104     EndDate DATE,
105     Notes TEXT,
```

Output

Action Output

#	Time	Action	Message
---	------	--------	---------

Object Info Session

Snipping Tool

Screenshot copied to clipboard and saved
Select here to mark up and share.

MySQL Workbench

DEMS Project x

File Edit View Query Database Server Tools Scripting Help

Navigator

SHEMAS

Filter objects

employees

sys

visa

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

No object selected

Limit to 2000 rows

```

19 PRIMARY KEY (user_id));
20
21 desc users;
22
23 INSERT INTO user(user_name, user_contact, user_address, user_id, PassportNumber, Nationality, DateOfBirth, Gender, Email)
24 VALUES('JAY', '1234567890', '121 main street', 101, 'S3U427', 'India', '2003-11-11', 'Male', 'jay@example.com'),
25 ('MOHAN', '1122334455', 'scheme no. 78', 102, 'LRV396', 'India', '1990-09-16', 'Male', 'mohan@example.com'),
26 ('RADHA', '8517955321', '11 bhawarkua', 103, 'KGF684', 'India', '2004-04-07', 'Female', 'radha@example.com'),
27 ('JOHN', '2314356588', '25 skyheights', 104, 'JFK320', 'UK', '2002-05-17', 'Male', 'john@example.com'),
28 ('SHIVI', '5564738285', 'Navlakha', 105, 'BVC154', 'India', '2002-07-07', 'Female', 'shivi@example.com'),
29 ('RAY', '7486256600', 'scheme 141', 106, 'NPM884', 'India', '2001-06-09', 'Male', 'ray@example.com'),
30 ('ROZY', '3354253435', 'lotus valley', 107, 'DAF768', 'Canada', '1996-10-20', 'Female', 'rozy@example.com'),
31 ('ZOEY', '9102037465', 'Raj palace', 108, 'RSE956', 'USA', '1998-08-10', 'Female', 'zoey@example.com'),
32 ('Pritam Lodhi', '2200110044', 'Chawani main street', 109, 'KGH432', 'INDIA', '2003-06-15', 'Male', 'pritam@example.com'),
33 ('Sunil Solanki', '9109100000', '5 GPO main street', 110, 'UNP141', 'UK', '1995-04-3', 'Male', 'sunil@example.com'),
34 ('John Doe', '1234567894', '123 Main St', 111, 'ABC123', 'USA', '1990-05-15', 'Male', 'john@example.com'),
35 ('Jane Smith', '9876543210', '456 Oak St', 112, 'XYZ456', 'Canada', '1985-10-20', 'Female', 'jane@example.com'),
36 ('Alice Johnson', '4567890123', '789 Elm St', 113, 'DEF789', 'UK', '1995-03-08', 'Female', 'alice@example.com'),
37 ('Ray Willson', '7634567890', '22 main st.', 114, 'PQR121', 'USA', '1992-03-25', 'Male', 'ray25@example.com'),
38 ('Reena Mourya', '9826612345', '138 GPO Main Road', 115, 'TUV112', 'India', '2000-04-16', 'Female', 'reena16@example.com'),
39 ('Jane Loui', '1724666305', '11A A.B Road', 116, 'BCD221', 'Canada', '2000-02-12', 'Male', 'jane12@example.com'),
40 ('Priyank Sharma', '3465772298', 'Block 14 Meera Road', 117, 'JKL312', 'India', '1990-12-12', 'Male', 'priyank@example.com'),
41 ('James James', '9876512345', '22 Skyheights', 118, 'GHI432', 'UK', '1987-10-18', 'Male', 'james@example.com'),
42 ('Daisy Merchant', '0453278611', 'Angels Colony', 119, 'RST001', 'UK', '2001-04-20', 'Female', 'daisy@example.com'),

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
---	------	--------	---------	------------------

Object Info Session

103°F Mostly sunny

Search

ENG IN 4:07 PM 5/12/2024

MySQL Workbench

DEMS Project x

File Edit View Query Database Server Tools Scripting Help

Navigator

SHEMAS

Filter objects

employees

sys

visa

Tables

Views

Stored Procedures

Functions

Administration Schemas

Information

No object selected

Limit to 2000 rows

```

58 desc visa_application;
59
60
61 INSERT INTO visa_application(user_id, appl_id, visa_country, visa_type, user_passport_no, status)
62 VALUES(101, 'PR100101', 'INDIA', 'Education', 'S3U427', 'Approved'),
63 (102, 'PR100102', 'INDIA', 'work', 'LRV396', 'Pending'),
64 (103, 'PR100103', 'INDIA', 'Education', 'KGF684', 'Approved'),
65 (104, 'PR100104', 'UK', 'Work', 'JFK320', 'Approved'),
66 (105, 'PR100105', 'India', 'Tourist', 'BVC154', 'Rejected'),
67 (106, 'PR100106', 'INDIA', 'Work', 'NPM884', 'Pending'),
68 (107, 'PR100107', 'Canada', 'Tourist', 'DAF768', 'Pending'),
69 (108, 'PR100108', 'USA', 'Education', 'RSE956', 'Approved'),
70 (109, 'PR100109', 'INDIA', 'Tourist', 'KGH432', 'Approved'),
71 (110, 'PR100110', 'UK', 'Work', 'UNP141', 'Rejected'),
72 (111, 'PR100111', 'USA', 'Tourist', 'ABC123', 'Pending'),
73 (112, 'PR100112', 'Canada', 'Tourist', 'XYZ456', 'Pending'),
74 (113, 'PR100113', 'UK', 'Tourist', 'DEF789', 'Approved'),
75 (114, 'PR100114', 'USA', 'Work', 'PQR121', 'Pending'),
76 (115, 'PR100115', 'India', 'Tourist', 'TUV112', 'Approved'),
77 (116, 'PR100116', 'Canada', 'Tourist', 'BCD221', 'Approved'),
78 (117, 'PR100117', 'India', 'Work', 'JKL312', 'Approved'),
79 (118, 'PR100118', 'UK', 'Work', 'GHI432', 'Pending'),
80 (119, 'PR100119', 'UK', 'Education', 'RST001', 'Pending'),
81 (120, 'PR100120', 'USA', 'Education', 'FDC453', 'Approved');

```

Output

Action Output

#	Time	Action	Message	Duration / Fetch
---	------	--------	---------	------------------

Object Info Session

103°F Mostly sunny

Search

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Screenshot of working Project (Min. 5)

MySQL 8.0 Command Line CLI X + v

```
mysql> desc user;
```

Field	Type	Null	Key	Default	Extra
user_name	varchar(50)	NO		NULL	
user_contact	varchar(20)	NO		NULL	
user_address	varchar(255)	NO		NULL	
user_id	int	NO	PRI	NULL	
PassportNumber	varchar(20)	YES		NULL	
Nationality	varchar(50)	YES		NULL	
DateOfBirth	date	YES		NULL	
Gender	varchar(10)	YES		NULL	
Email	varchar(100)	YES		NULL	

9 rows in set (0.01 sec)

```
mysql> select*from user;
```

user_name	user_contact	user_address	user_id	PassportNumber	Nationality	DateOfBirth	Gender	Email
JAY	1234567890	121 main street	101	SJU427	India	2003-11-11	Male	jay@example.com
MOHAN	1122334455	scheme no. 78	102	LRV396	India	1990-09-16	Male	mohan@example.com
RADHA	8517955321	11 bhawarkua	103	KGF684	India	2004-04-07	Female	radha@example.com
JOHN	2314356588	25 skyheights	104	JFK320	UK	2002-05-17	Male	john@example.com
SHIVI	5564738285	Navlakha	105	BVC154	India	2002-07-07	Female	shivi@example.com
RAY	7486256600	scheme 141	106	NPM884	India	2001-06-09	Male	ray@example.com
ROZY	3354253435	lotus valley	107	DAF768	Canada	1996-10-20	Female	rozy@example.com
ZOEY	9102837465	Raj palace	108	RSE956	USA	1998-08-18	Female	zoey@example.com
Pritam Lodhi	2200110044	Chawani main street	109	KGH432	INDIA	2003-06-15	Male	pritam@example.com
Sunil Solanki	9109100000	5 GPO main street	110	UNP141	UK	1995-04-03	Male	sunil@example.com
John Doe	1234567894	123 Main St	111	ABC123	USA	1990-05-15	Male	john@example.com
Jane Smith	9876543210	456 Oak St	112	XYZ456	Canada	1985-10-20	Female	jane@example.com
Alice Johnson	4567890123	789 Elm St	113	DEF789	UK	1995-03-08	Female	alice@example.com
Ray Willson	7634567890	22 main st.	114	PQR121	USA	1992-03-25	Male	ray25@example.com
Reena Mourya	9826612345	13B GPO Main Road	115	TUV112	India	2000-04-16	Female	reena16@example.com
Jane Loui	1724666385	11A A.B Road	116	BCD221	Canada	2000-02-12	Male	Jane12@example.com
Priyank Sharma	3465772298	Block 14 Meera Road	117	JKL312	India	1990-12-12	Male	priyank@example.com
James James	9876512345	22 Skyheights	118	GHI432	UK	1987-10-18	Male	james@example.com
Daisy Merchant	0453278611	Angels Colony	119	RST001	UK	2001-04-20	Female	daisy@example.com
Henry Gill	8856934245	56 Main Road	120	FDC453	USA	1999-08-06	Male	henry6@example.com

103°F Mostly sunny

Search

ENG IN 5:11 PM 5/12/2024

```
mysql> select*from user;
```

user_name	user_contact	user_address	user_id	PassportNumber	Nationality	DateOfBirth	Gender	Email
JAY	1234567890	121 main street	101	SJU427	India	2003-11-11	Male	jay@example.com
MOHAN	1122334455	scheme no. 78	102	LRV396	India	1990-09-16	Male	mohan@example.com
RADHA	8517955321	11 bhawarkua	103	HGF684	India	2004-04-07	Female	radha@example.com
JOHN	2314356588	25 skyheights	104	JFK320	UK	2002-05-17	Male	john@example.com
SHIVI	5564738285	Navlakha	105	BVC154	India	2002-07-07	Female	shivi@example.com
RAY	7486256600	scheme 141	106	NPM884	India	2001-06-09	Male	ray@example.com
ROZY	3354253435	lotus valley	107	DAF768	Canada	1996-10-20	Female	rozy@example.com
ZOEY	9102837465	Raj palace	108	RSE956	USA	1998-08-18	Female	zoey@example.com
Pritam Lodhi	2200110044	Chawani main street	109	HGH432	INDIA	2003-06-15	Male	pritam@example.com
Sunil Solanki	9109100000	5 GPO main street	110	UNP141	UK	1995-04-03	Male	sunil@example.com
John Doe	1234567894	123 Main St	111	ABC123	USA	1990-05-15	Male	john@example.com
Jane Smith	9876543210	456 Oak St	112	XYZ456	Canada	1985-10-20	Female	jane@example.com
Alice Johnson	4567890123	789 Elm St	113	DEF789	UK	1995-03-08	Female	alice@example.com
Ray Willson	7634567890	22 main st.	114	PQR121	USA	1992-03-25	Male	ray25@example.com
Reena Mourya	9826612345	138 GPO Main Road	115	TUV112	India	2000-04-16	Female	reena16@example.com
Jane Loui	1724666385	11A A.B Road	116	BCD221	Canada	2000-02-12	Male	Jane12@example.com
Priyank Sharma	3465772298	Block 14 Neera Road	117	JKL312	India	1990-12-12	Male	priyank@example.com
James James	9876512345	22 Skyheights	118	GHI432	UK	1987-10-18	Male	james@example.com
Daisy Merchant	0453278611	Angels Colony	119	RST001	UK	2001-04-20	Female	daisy@example.com
Henry Gill	8856934245	56 Main Road	120	FDC453	USA	1999-08-06	Male	henry6@example.com



```
mysql> select * from executive;
```

exe_id	exe_name	exe_address	exe_contact_no
1	John Doe	123 Executive Street	123-456-7890
2	Jane Smith	456 Leadership Avenue	987-654-3210
3	Michael Johnson	789 Management Road	555-123-4567
4	Emily Brown	321 Director Lane	888-999-0000
5	David Wilson	555 Executive Court	777-888-9999

```
5 rows in set (0.01 sec)
```

```
mysql> select * from visa_renewal_application;
```

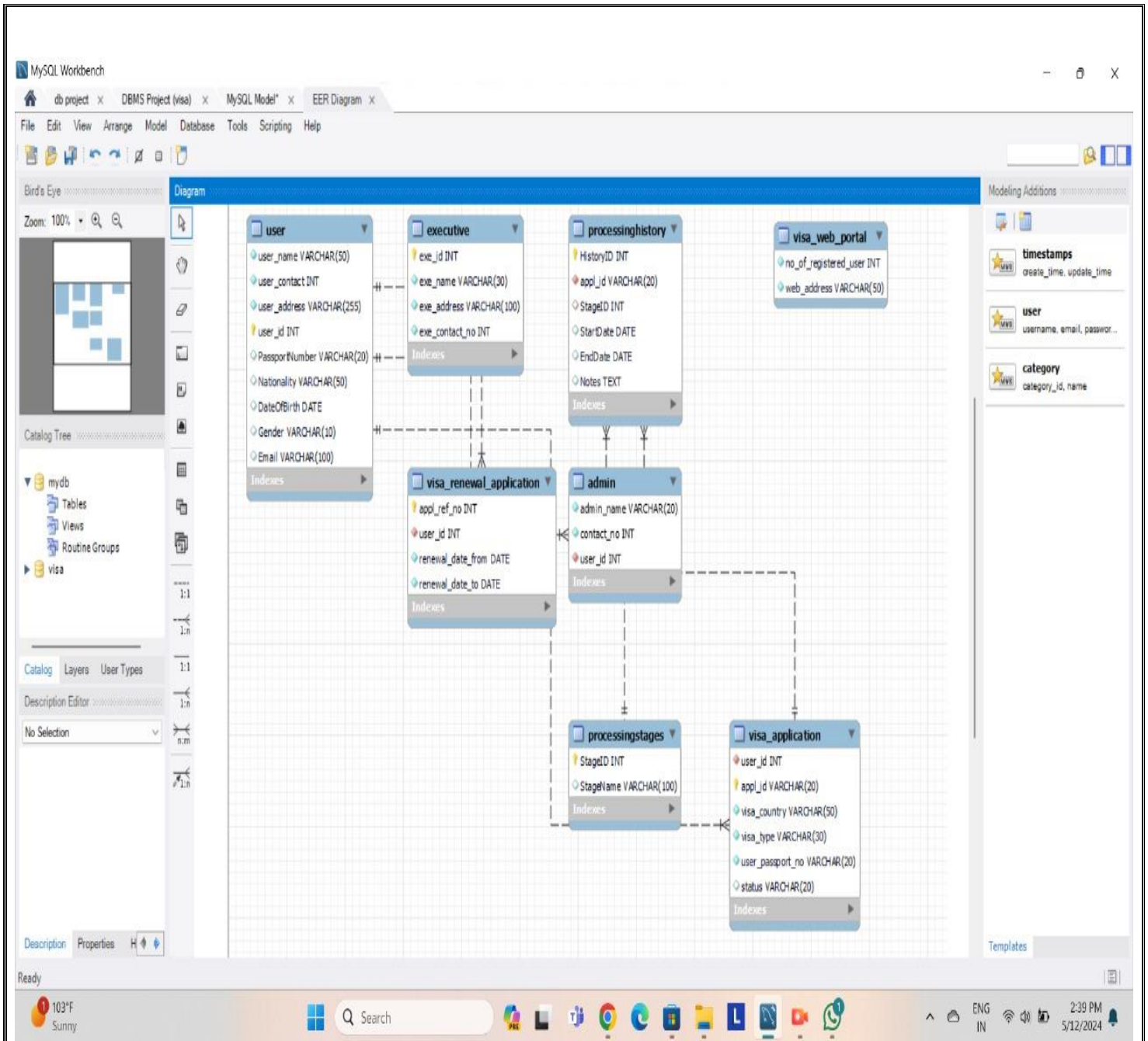
appl_ref_no	user_id	renewal_date_from	renewal_date_to
1	101	2024-05-10	2024-06-10
2	102	2024-06-15	2024-07-15
3	103	2024-07-01	2024-08-01
4	104	2024-08-15	2024-09-15
5	105	2024-09-20	2024-10-20
6	106	2024-10-25	2024-11-25
7	107	2025-01-15	2025-02-15
8	108	2025-02-20	2025-03-20
9	109	2025-03-25	2025-04-25
10	110	2025-04-30	2025-05-30

```
10 rows in set (0.01 sec)
```

```
mysql> select * from admin;
```

admin_name	contact_no	user_id
Admin1	123-456-7890	101
Admin2	987-654-3210	102
Admin3	555-123-4567	103
Admin4	999-888-7777	104
Admin5	444-333-2222	105
Admin6	111-222-3333	106
Admin7	666-777-8888	107
Admin8	888-999-0000	108
Admin9	333-444-5555	109
Admin10	555-666-7777	110

```
10 rows in set (0.02 sec)
```



References

- W3Schools (<https://www.w3schools.com/>)
- Youtube 1.(<https://www.youtube.com/watch?v=dl00fOOYLOM>)
2.(<https://www.youtube.com/watch?v=hlGoQC332VM>)
- javatpoint (<https://www.javatpoint.com/>)